

CS 352 Internet Technology

Project 1 Report Questions: TCP Socket Programming, Message Validation, and Packet Capture

- Released: September 15, 2026; Due: October 6, 2026
- Submit as report.pdf inside the team ZIP

Instructions

Create a typed report that answers every item below. Use frame numbers from your own captures; do not copy sample frame numbers. Wireshark relative sequence numbers are acceptable; state whether you used relative or absolute numbering. Refer to packets by frame number, and if a field is not present, explain why rather than leaving the answer blank. Include legible screenshots only where they help support an answer.

Part A. Team and experiment setup

1. Team members (names and netids):

2. Assigned TCP server port: _____

3. Client/server host names or IP addresses. Capture 1 (same machine):

_____ Capture 2 (different machines): _____

4. Collaboration and resources consulted. State who or what you consulted and the specific aspect of the project:

Part B. Protocol basics

1. What transport-layer protocol is used in your capture? _____

2. What port number did the server use? _____

3. What ephemeral port did the client use? (in part1, and part2) _____

4. What interface was used when you ran the C&S on the same machine (rlab5)?

5. What interface was used when you ran the C&S on different machines (client on rlab5)? _____

Part C. One application request and response

Select one complete client request and its corresponding server response.

1. Request frame number: _____ Request tcp.seq: _____ Request tcp.len: _____

2. Decoded request record: _____

3. Response frame number: _____ Response tcp.len: _____

4. Decoded response record: _____

5. Identify the packet that acknowledges the request payload. ACK frame: _____

Observed ACK number: _____ Request seq + request tcp.len: _____ Match
(yes/no): _____

6. Show the numerical calculation:

Part D. Application validation

1. For the selected response, compare the returned line number with the request line number. Request line: _____ Returned line: _____ Line-number check (pass/fail): _____
2. Verify the returned message length. Reported length: _____ Actual UTF-8 payload bytes: _____ Length check (pass/fail): _____ Output record written (full or ERROR): _____
3. Did the capture contain any response with a deliberately incorrect reported length? If yes, give its frame number and both lengths. If no, state that no fault occurred in this run.

Part E. Traffic totals

1. Capture 1 (same machine). Client to server payload bytes: _____ Server to client payload bytes: _____
2. Capture 2 (different machines). Client to server payload bytes: _____ Server to client payload bytes: _____
3. State the Wireshark or tshark filter or method used to calculate the totals:

4. Identify the packet with the largest TCP payload in either capture. Capture: _____
Frame: _____ Direction: _____ tcp.len: _____

Final checklist

- Every value comes from your own capture; no sample frame numbers are copied.
- Your ACK arithmetic is shown, not only stated.
- Your report identifies whether the client writes a full response record or an ERROR record.
- The final file is named report.pdf and is included in the submission ZIP.